Journal Title Here, 2024, 1–2
doi: DOI HERE
Advance Access Publication Date: Day Month Year
Paper

More cells, more doublets in sample-barcoded single-cell data

George Howitt,[1,2] Gunjan Dixit,[1,2] Rotem Aharon,[1,2] Victoria Streeton-Cook,[1] Ling Ling,[3] Peter F. Hickey,[3,4] Daniela Amann-Zalcenstein,[3,4] Liam Gubbels,[5] Shivanthan Shanthikumar,[5,6,7] Sarath Ranganathan,[5,6,7] Melanie R. Neeland,[5,6] Jovana Maksimovic[1,2] and Alicia Oshlack[1,2,8,*]

[1]Computational Biology Program, Peter MacCallum Cancer Centre, Parkville, VIC, Australia, [2]Sir Peter MacCallum Department of Oncology, University of Melbourne, Parkville, VIC, Australia, [3]The Walter and Eliza Hall Institute of Medical Research, Parkville, VIC 3052, Australia, [4]Department of Medical Biology, University of Melbourne, Parkville, VIC, Australia, [5]Respiratory Diseases, Murdoch Children's Research Institute, Parkville, VIC, Australia, [6]Respiratory and Sleep Medicine, Royal Children's Hospital, Parkville, VIC, Australia, [7]Department of Paediatrics, University of Melbourne, Parkville, VIC, Australia and [8]School of Mathematics and Statistics, University of Melbourne, Parkville, VIC, Australia

[*]alicia.oshlack@petermac.org

FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

Abstract

Withdrawal statement

In our initial submission of this paper, we proposed a combinatoric model for the probability of finding doublets in sample-barcoded single-cell RNA-sequencing data. This model predicts a doublet rate higher than the rate included in the documentation for the 10X Flex protocol. This was motivated by our use of doublet-finding software, scDblFinder, on experimental data using the Flex protocol, which identified several times more doublets than we expected based on the documentation. Our original model produced better agreement with the results of scDblFinder in 9 of our own data sets as well as one public dataset made available by 10X.

During revisions, however, we performed Monte Carlo simulations of doublet formation that were inconsistent with the results of our model, and much closer to the predictions from the documentation. This prompted us to reanalyse the assumptions of our model. We go into the mathematical details below, but in summary, our initial assumptions over-predicted the doublet fraction, and our revised model is now consistent with both the Monte Carlo simulations and the predictions from the Flex documentation. We therefore speculate that the unexpectedly high doublet fractions observed in the experimental data might be due to some combination of experimental conditions and software performance and therefore withdraw this paper.

Estimating the doublet fraction in Flex data

We have written code to simulate the formation of doublets in Flex data. The simulation follows a simple algorithm:

- Generate a vector of droplet occupancies through pseudo-random sampling of a zero-truncated Poisson distribution.
- Populate the droplets with cells by random sampling with replacement from a list of barcodes (samples).
- For each droplet:
  - Compute the number of doublets by counting the number of samples with more than one cell in that droplet.
  - Compute the apparent number of cells in the droplet by counting the number of unique samples present.
  - Compute the doublet fraction by summing the number of doublets divided by the number of apparent cells across all droplets.

As a test of the simulation code, we aimed to reproduce the results of our combinatoric model in the case where each sample is equally likely. In fact, the simulations predicted a much lower doublet fraction than the model, shown in Figure A1

To understand this discrepancy, we then looked at the assumptions of the combinatoric model. Equation (1) in the paper states correctly that for a droplet containing k cells drawn from a pool with n unique samples, the number of distinguishable configurations of cells within droplets is the number of combinations with repetition allowed,

$$M_k(n) = \frac{(n+k-1)!}{(n-1)!k!} \quad .$$

We then assumed that each of these combinations is equally probable. However, this assumption is incorrect. Consider a droplet with two cells drawn from samples A and B. There are four possible outcomes: AA, AB, BA, BB,

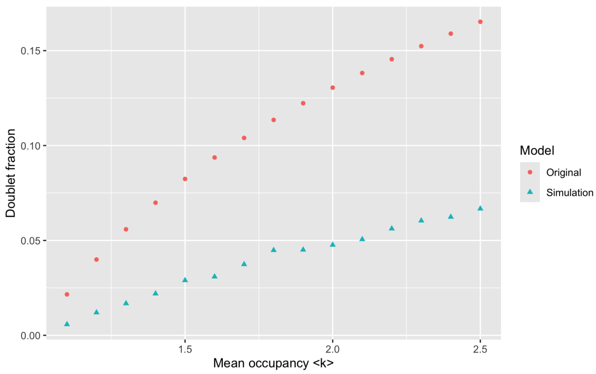

Fig. A1: doublet fraction vs mean occupancy for the combinatoric model in the original draft of the paper (red circles), and from simulations (teal triangles).

each with equal probability in a random sampling. In a Flex experiment, the outcomes AB and BA are indistinguishable, but the probability of observing one cell from sample A and one from sample B is double that of observing two cells from sample A.

Taking this into account, the number of configurations of k cells drawn from n samples that have equal probability of occurrence is instead the number of permutations with repetition allowed,

$$M_k(n) = n^k \quad .$$

To determine the fraction of doublets, we can use the same concept of partitions as in the original version of the model. For each partition of k we compute the doublet fraction of the partition and the number of configurations corresponding to the partition. The doublet fraction of droplets with occupancy k is then the sum over partitions of the partition doublet fraction multiplied by the number of partition configurations, divided by the total number of configurations. Equation(4) in the original manuscript gives a formula for the number of unique configurations for each partition:

$$C_i(n,k) = \prod_{j=1}^{k} \binom{n-d_j}{m_j} \quad ,$$

where i denotes a partition of k, m_j is the number of samples with j repeats in the partition, and d_j is a counting variable. If we instead count the unique configurations as permutations rather than combinations, equation (4) needs to be multiplied by an additional factor, with is the number of k-permutations of the multiset represented by the partition:

$$C_i(n,k) = \frac{k!}{\prod_{j=1}^{k} m_j!} \prod_{j=1}^{k} \binom{n-d_j}{m_j} \quad .$$

With this correction in place, we find that the predictions of the model now match the simulations, and also hew closely to the predictions of the 10X documentation, as shown in Figure A2.

Code to reproduce the figures in this statement are available at `https://github.com/Oshlack/flex-doublets`.

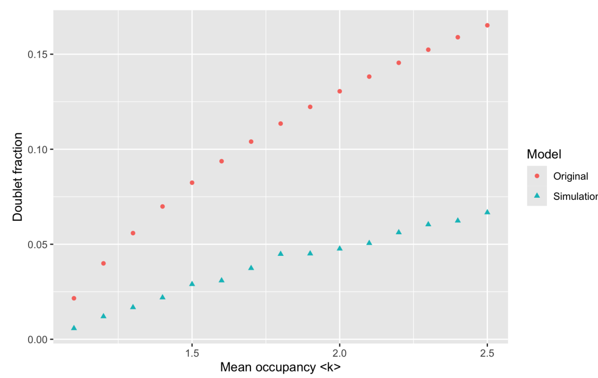

Fig. A2: Doublet fraction vs mean droplet occupancy. Red squares indicate the values from the combinatoric model in the original version of the paper. Blue squares indicate the values from simulation. Green squares indicate the values from the model with the correction applied. Black circles are the predictions from the Flex documentation for the earlyAIR and PBMC data sets, black triangles are the values from scDblFinder.